

SMART INDIA HACKATHON 2026

Problem Statement ID - 26131

Problem Statement Title - Early detection and management of crop diseases and pest infestations.

Theme - Agriculture, Food-Tech & Rural Development

PS Category - Software

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TerraSense AI Maharashtra - Project Overview

Smart India Hackathon 2026

Problem Statement PS26131 - Early detection and management of crop diseases and pest infestations.

Government of Maharashtra · Maharashtra State Innovation Society (Dept. of Skills, Employment, Entrepreneurship and Innovation)

Category: Software

Theme: Agriculture, Food-Tech & Rural Development

Welcome to TerraSense AI, we have built a **proactive, offline crop-health platform and mobile app designed for farmers and agricultural officers**, focused on different crops (potatoes for the prototype now) with an **in-development research lab for autonomous field robotics**. It pairs on-device camera scanning with weather-driven and data forecasting, safety-gated multilingual advisories with explanations, cross-farm outbreak intelligence, and a geospatial dashboard. Every requirement from the problem statement is fully implemented and directly traceable to our code (verified in **docs/PS_TRACEABILITY.md** of our GitHub repository).

1. The Problem We Are Solving

1.1 The problem, in the field

Farming is incredibly time-sensitive, and current tools leave farmers with major blind spots:

- **Reactive diagnosis:** Farmers usually only detect a disease when visible damage has already ruined the foliage. By then, yield loss is inevitable.
- **Overstretched extension staff:** Dedicated agricultural officers cover massive geographical territories, meaning on-the-ground expert diagnosis cannot reach every village immediately.
- **Disconnected agronomic signals:** Microclimate shifts, soil moisture, and neighbourhood outbreak history rarely translate into clear, farm-level daily advice.

- **The chemical cycle:** Fear and delayed diagnosis push farmers into panic-spraying broad-spectrum pesticides, raising input costs, creating chemical resistance, and contaminating soil and runoff.

1.2 Official problem statement (PS26131)

Farmers often recognise crop diseases or pest infestations only after visible damage has spread. Extension staff may cover large areas, while laboratory diagnosis and expert advice may not be immediately available. Weather, crop stage, variety, soil condition and local pest history influence risk, but these inputs are rarely combined into actionable farm-level alerts. Incorrect diagnosis may lead to delayed treatment, excessive or inappropriate pesticide use, increased cultivation cost, residue concerns and yield loss. The challenge is to provide timely, reliable and locally relevant detection, forecasting and management support.

Expected solution (official): A farmer- and extension-worker-friendly crop-health system supporting image-based symptom identification, pest-trap or sensor inputs, weather-based risk forecasting, geospatial hotspot mapping, expert validation, and multilingual advisories. It should recommend integrated pest and disease management (IPDM) actions, safe input usage, referral to extension officers or laboratories, and follow-up monitoring; learn from field confirmations; and provide dashboards for agriculture officials.

2. The Solution Architecture

TerraSense AI Maharashtra delivers a closed-loop platform: **Forecast → Triage → Detect → Verify → Monitor → Learn.**

For Farmers

- **The "Should I walk my field today?" Home Landing:** Instead of facing a blank camera screen, farmers open to a single categorical traffic light (**Calm / Watch / Act**) with plain, simple language instructions in Marathi, Hindi, Bengali, or English.
- **Proactive Risk Alerts:** Weather-driven and pheromone trap telemetry data driven epidemiological modelling that alerts farmers days before pathogen spores visibly manifest.

- **On-Device Offline Live Video Scanner:** Edge AI camera scanning running inside the mobile app fully offline and giving diagnosis and advisories (with explainable AI) at zero per-frame server cost, operational deep in rural dead zones without an internet connection.
- **Safe, Localized Advisories with XAI:** Non-hallucinatory Integrated Pest & Disease Management (IPDM) guidance, along with explanations, with an explicit "Do Not Spray" directive for healthy crops.
- **Follow-Up & Resistance Tracking:** Systematic check-ins on treatment efficacy that escalate repeated failures to human laboratories rather than prescribing heavier chemical doses.

For Extension Officers & State Officials

- **Explainable Evidence Auditing:** Officers can click through any automated risk score to see the exact meteorological formulas and weightings that triggered the alert.
- **Geospatial Hotspot Mapping:** Dual-layer surveillance featuring a cell-by-cell explainability grid alongside village-level kernel density heatmaps.
- **Pest-Trap Sensor Ingestion:** Direct telemetry ingestion from in-field pheromone traps feeding tuber-moth population counts into the map.
- **Human-in-the-Loop Review Queue:** Agricultural officers validate AI detections, feeding verified local cases back into the training lifecycle.

3. The CropRow Lab - In-Development Vision Research (Beta)

- **What it is:** A separate, deliberately walled-off research module located under the navigation group "**In the lab · beta**".
- **Scope Discipline:** It answers a distinct robotics question: "**Where are the healthy and unhealthy crop plants positioned for a cultivator-mounted camera?**". For now, it just acts as a two-class plant localizer, generating zero database cases, zero chemical advisories, and zero patient records.

- **Dual-Stream Processing:** Supports live real-time field camera feeds alongside uploaded video clips, rendering dynamic bounding boxes on active footage.
- **Future plan and current limitations:** We plan to use this feature to detect diseased crops and classify them into healthy or unhealthy instantly (within a fraction of seconds) for the cultivator to discard/remove the unhealthy ones. This has to be achieved with high speed, high recall and high accuracy. Currently we could not find the appropriate datasets, hence we had to limit the prototype, but we are actively working on it.

4. What's Under the Hood? (Our Tech Stack)

Client Interfaces (Frontend)

- **React (Vite):** Core architecture for Farmer Home, Officer Review Queue, Risk Surfaces, and the CropRow Lab (both for web-app and mobile-app).
- **onnxruntime-web (WebAssembly):** Executes client-side computer vision models entirely within browser memory on standard phone CPUs and it runs fully offline after a one-time download.
- **Leaflet & Mapbox GL:** Geospatial map rendering, clustering, and kernel density heatmap generation.
- **Recharts:** Epidemiological trendlines, multi-day humidity duration charts, and trap count curves.

Application & Inference Serving (Backend)

- **FastAPI:** High-performance asynchronous REST API.
- **Shell & Backend for Mobile App:** Flutter, Dart, Java, Kotlin/ Gradle.
- **SQLite:** Structured persistence for case histories, trap readings, audit logs, and expert review queues.
- **Uvicorn:** Production ASGI web serving layer.
- **Model Hosting:** Static Object Storage (S3, GitHub Pages, etc.)

Computer Vision Models

- **YOLOv8s (Potato Pathology):** Visual disease detector trained on combined laboratory and field splits, exported to ONNX for CPU-only serving.
- **YOLOv11n-CropRow (Cultivator Guidance):** Two-class plant localization model wired to plantTracker.js.
- **Ultralytics & PyTorch:** Cloud-isolated training environment.

Agronomic Modeling & Advisory

- **Classical Agronomy:** Hardcoded mathematical routines for Smith Period, Beaumont Period, and TOMCAST DSV.
- **XGBoost + TreeSHAP:** Auxiliary risk refinement explainable AI layer with hardcoded ± 0.20 safety limits.
- **LangGraph RAG Engine:** Multi-stage advisory pipeline (retrieve → compose → safety gate → localize).
- **ChromaDB & BM25:** Hybrid semantic vector and keyword search accessing human-audited markdown knowledge bases.
- **Deterministic Localization:** Static catalogue covering 49 messages across Marathi, Hindi, Bengali, and English.

5. Our Mobile App:

Android app for the farmer. The whole farmer loop is weather risk forecast → photo → diagnosis → triage → treatment advice → follow-up. This runs on the handset with the radio off. A connection is needed exactly twice: once on first launch to download a crop model, and afterwards only for cross-farm data that by definition lives on other people's devices.

What features work Offline vs what online is given below:

Offline	Online
• Weather feed • Risk models • Geo / hotspot cells • Triage safety gate • Class taxonomy, 4 languages • Photo Diagnosis • Live camera scanner • Lab scanners (crop row, crop health) • Advisory / RAG • Dose tables • Follow-up scheduling • Officer screens on demo data	• One-time downloading a crop pack • Cross-farm outbreak pressure (hotspot map) • Officer dashboard, review queue • Chat bot • Real-time weather forecast pre-fetch

6. How It Actually Works

We wanted the flow to be completely natural for our farmers, but incredibly robust behind the scenes:

- Farmers use the live scanner, upload a photo, or the system runs a proactive weather check when there is no image at all.
- The backend assesses the image and weather data, routing every single case through a single safety triage gate in pipeline.py. This ensures a weather alert can never bypass a safety rule that a photo case obeys.
- The system retrieves guidance from a human-reviewed knowledge base to generate a localized advisory, returning the exact KB sections it drew from as citations for transparency.
- To prevent false alarms, the live camera filters out blurry frames (using variance-of-Laplacian blur and exposure scoring) and requires a consensus of multiple clear frames-by default, 6 out of 10 good frames at $\geq 55\%$ mean confidence-before issuing a verdict.
- Nothing is stored until the farmer presses Accept, which sends that exact frame through the full /detect pipeline-resulting in the same advisory, same triage, and same follow-up as a photo upload. Discarded scans leave absolutely no record.

- Later, an officer reviews the case, confirming or correcting the diagnosis. A confirmed case safely becomes a training sample, and a failed follow-up auto-escalates the case.

7. Competitive Advantages & Unique Selling Propositions

- **Precaution is better than cure:** Most crop tools only react after the damage shows up. TerraSense AI works the other way around. Using published agronomic models like the Smith and Beaumont Periods and TOMCAST, it reads the weather building up around a field and warns the farmer that blight is coming days before the first spot appears on a leaf. The alert is not a black box either, since every model that fired and every reason behind the score is shown in plain language. It is the difference between treating a disease and quietly preventing it, giving the farmer a head start when it still matters.
- **Hyperlocal microclimate sensing (experimental):** We turn the phone's live camera into a passive, in-field airflow proxy. Classical pixel math cancels camera shake and reads the independent leaf motion that survives, grading canopy airflow as still / light / breezy-a crucial signal a distant weather station cannot see. Still air means dew lingers, stretching out leaf wetness, which is exactly the input the Smith Period (≥ 11 h) and TOMCAST blight models read. Paired with soil-drainage modelling, this localises forecasting to the exact plant in front of the farmer. A strict guardrail keeps it safe: a breeze may raise concern but can never "un-fire" a blight rule that already met the observed humidity.
- **Offline Ready:** We run live camera scanning and multilingual advisories (Marathi, Hindi, Bengali) totally offline without expensive API calls. The WASM runtime is served from the app's own origin-not a CDN-precisely so our offline claim is real. **Even the mobile app runs all the models and all diagnosis offline with just a one-time download of a light-weight model of your preferred crop.**
- **Honest AI:** We explicitly separate broken components (degraded) from deliberate design limitations (by design) on the dashboard, so a judge, officer, or teammate never has to guess whether a number came from a real model or a fallback. We never fabricate a result just to look complete.

- **A learning flywheel that improves in the field:** Every case an officer confirms or corrects in the review queue becomes a labelled Training Sample. Our script (ml/export_feedback.py) packages these for retraining and warns us when a batch is too small or skewed to one district. The retrain is a deliberate, human-in-the-loop step, not blind auto-training- because retraining on unexamined field data is how a model quietly degrades over time.
- **Two decoders, one behaviour:** Our browser (JavaScript) and server (Python) YOLO decoders are tested against each other case-for-case, ensuring both produce the exact identical bounding box.

8. Target Customers and Societal Impact

Designed with empathy for farmers, local extension workers, and state agriculture officials.

The Impact We Aim For:

- Reduces crop loss by predicting threats early. For instance, a fired "Smith Period" is an infection event, not a visible one, so our warning arrives before the actual symptoms do.
- Promotes targeted pesticide use by specifically advising farmers when not to spray healthy crops, withholding dose tables when our confidence is low, and routing repeated treatment failures to a lab instead of just suggesting another spray.
- Empowers officials to deploy their staff efficiently based on verified hotspot data, with confirmed cases weighted heavily above unverified model outputs.
- Improves surveillance coverage-every single photo taken becomes a georeferenced case, and our pest-trap ingestion adds a powerful second signal stream.

9. What's Next? (Future Scope & Scalability)

- **More crops and classes:** Our architecture generalizes beautifully. Adding a new crop is simply a dataset-and-retrain task—we just extend `ml/data.yaml`, `taxonomy.py`, and the knowledge base. Tomato and bell pepper are the natural next crops as they belong to the same nightshade family, sharing vulnerabilities to late blight and utilizing similar forecasting metrics like the Smith Period.
- **Dedicated field metric:** We plan to expand the field-image dataset to produce a standalone field-condition validation split, ensuring **accuracy, speed and recall** is always quoted on real phone photos.
- **Activate the XGBoost + SHAP risk layer:** We will turn this on once enough real Indian outbreak data accrues. Our `scripts/export_risk_dataset.py` already builds a leakage-safe training set from confirmed cases. The layer is capped at ± 0.20 so it can refine, but never override, a validated agronomic rule.
- **Real weather history:** We plan to backfill the Smith Period window from a paid or archival weather source instead of relying on the system's own accumulating cache.
- **CropRow Disease-detection feature:** The beta crop-row lab already localizes plants and counts healthy and unhealthy plants without double-counting from a moving camera; the next step is mounting it on a real cultivator to steer inter-row weeding and detect the unhealthy plants too this time.

10. Revenue System

TerraSense AI is primarily designed strictly as a public good for the Government of Maharashtra for the Smart India Hackathon. It features no commercial subscription or monetization model whatsoever. Our entire focus is on minimizing operational costs (utilizing free open-source APIs and offline tools) while generating massive societal value by saving crops and empowering our rural communities.

Our repository: https://github.com/aliviahossain/Crop_detection_management

Download our APK: https://aliviahossain.github.io/Crop_detection_management/

Deployed Application: <https://cropguard-frontend-rhzv.onrender.com/>

11. Competitor Analysis Report:

Market Leaders vs. TerraSense AI Maharashtra

11.1 Direct Competitor Landscape & Verified Benchmarks

- **Plantix (Global Benchmark):** Over 10 million downloads and ~8 million annual active users. Processes 26,000+ daily scans across 30 crops detecting 780+ diseases in 1.7 seconds via cloud CNNs.
Limitation: Strictly reactive (detects after leaf tissue is dead), cloud-dependent, no canopy airflow sensing, and no pre-symptom epidemiological forecasting.
- **MahaVISTAAR AI (Maharashtra State Voice AI):** 2.5 million+ farmers, 20,000+ daily queries across 19 languages/dialects (including tribal Bhili). Operates without smartphones via toll-free voice calls.
Limitation: Conversational only; zero visual computer vision, no row tracking, and no localized disease spread modeling.
- **BharatAgri & AgroStar (Commercial Advisory & E-Commerce):** 5.5M+ and 10M+ downloads respectively. Offer crop calendars, agronomist consultations, and pest identification. **Limitation:** Commercial conflict of interest-revenue is tied directly to selling agrochemicals, creating an inherent incentive to prescribe commercial sprays rather than integrated pest management (IPM).
- **Agrio (Precision Agriculture):** ~800,000 global users. Employs satellite remote sensing (NDVI) and macro weather spore-spread models.
Limitation: Expensive enterprise tier, high data bandwidth requirement, complex UX, and lacks local Indian dialect support or hyper-local on-device canopy sensing.
- **CROPSAP Maharashtra (Legacy State Surveillance):** Statewide government initiative relying on physical field scouts logging trap counts manually to web dashboards for scientist review. **Limitation:** Heavy human dependency, slow turnaround times, and coarse village-level advisories rather than immediate farm-level guidance.

11.2 Expanded Systematic Feature Comparison Matrix

Architectural Feature	TerraSense AI Maharashtra	Plantix	Agrio	AgroStar / BharatAgri	MahaVISTAAR / CROPSAP
First-Screen Interaction	"Should I walk my field today?" Traffic Light (Calm / Watch / Act)	Blank camera viewfinder / Upload form	Farm field polygon / satellite map	E-commerce banner / advisory chat feed	Voice call prompt / Text bulletin boards
Operational Paradigm	Pre-Symptomatic Forecast + Real-Time Video Consensus	Reactive (Post-symptom still photo)	Macro-satellite forecast + reactive photos	Reactive (Post-symptom photo upload)	Periodic manual field scout reports
Disease Modeling Architecture	Agronomic Science First, ML Second (Smith, Beaumont, TOMCAST, DD + SHAP)	Black-box Deep CNN classification	Cloud-based predictive weather models	Human agronomist / Basic rules	Manual scientist review of pest traps
Explainability Layer	Full Evidence Matrix (Shows exact rule triggers & modifier weights)	Opaque confidence percentage	Black-box risk index	Informal chat explanation	Published PDF / text advisories
Micro-Canopy Sensing	Camera Pixel Transition (Zero-hardware local wind & wetness)	Generalized weather APIs (20–30 km away)	Satellite & regional weather grid	3rd-party weather APIs (IMD/Accu Weather)	Manual field rain gauges & pheromone traps
Spatial Outbreak Pressure	Cross-Farm Aggregation (Privacy-preserving 5	Regional heatmap (Cloud aggregate)	Satellite macro-spread tracking	Isolated user logs	Village-level scout aggregates

	km / 15 km radius)				
Network & Edge Capability	100% Offline Edge-AI (On-device ONNX runtime)	Requires internet for photo upload	Requires internet for satellite /inference	Requires active data connection	Cellular voice / Web portal
Safety & Chemical Stewardship	Safety-Gated Withholding (Blocks dosage on low confidence/resistance)	Recommends commercial pesticides	Recommends spray windows	Commercial bias: Drives pesticide sales	Non-commercial public advisories
Ecosystem Integration	Dual-Sided Unified (Farmer app + State Officer GIS Portal)	B2C Farmer-only (Paid enterprise APIs)	Enterprise B2B / Large farm focus	B2C Farmer marketplace	B2G Government portal only
Model Evolution	Human-in-the-Loop (HITL) (Officer-validated review queue)	Centralized batch training on uploads	Centralized cloud updates	Manual agronomist adjustments	Manual periodic advisory updates
Robotics & Spatial Lab	CropRow Lab (Beta) (IoU plant tracking and health detection)	None	None	None	None

Farmer App Adoption & Reach

- **Overall Usage:** Across India, approximately 10% to 15% of smartphone-owning farmers actively use agriculture-specific mobile applications.
- **Device Penetration:** Around 70% of educated Indian farmers possess a smartphone, but only 10% to 15% routinely engage with dedicated agri-apps. The remainder primarily consume farming information via WhatsApp peer groups and YouTube.

- **Maharashtra's Leadership:** Maharashtra has emerged as a primary hub for digital onboarding, supported by programs like the MahaAgri-AI Policy (2025–2029).
- **MahaVISTAAR AI:** The state's bilingual advisory platform has successfully onboarded over 3 million (30 lakh) farmers, providing them access to real-time weather, pest management, and advisory services.
- **Digital Registries:** Under the national AgriStack initiative, millions of land-verified IDs have been generated in Maharashtra to streamline crop verification, credit access, and insurance.

Productivity & Income Growth Rates

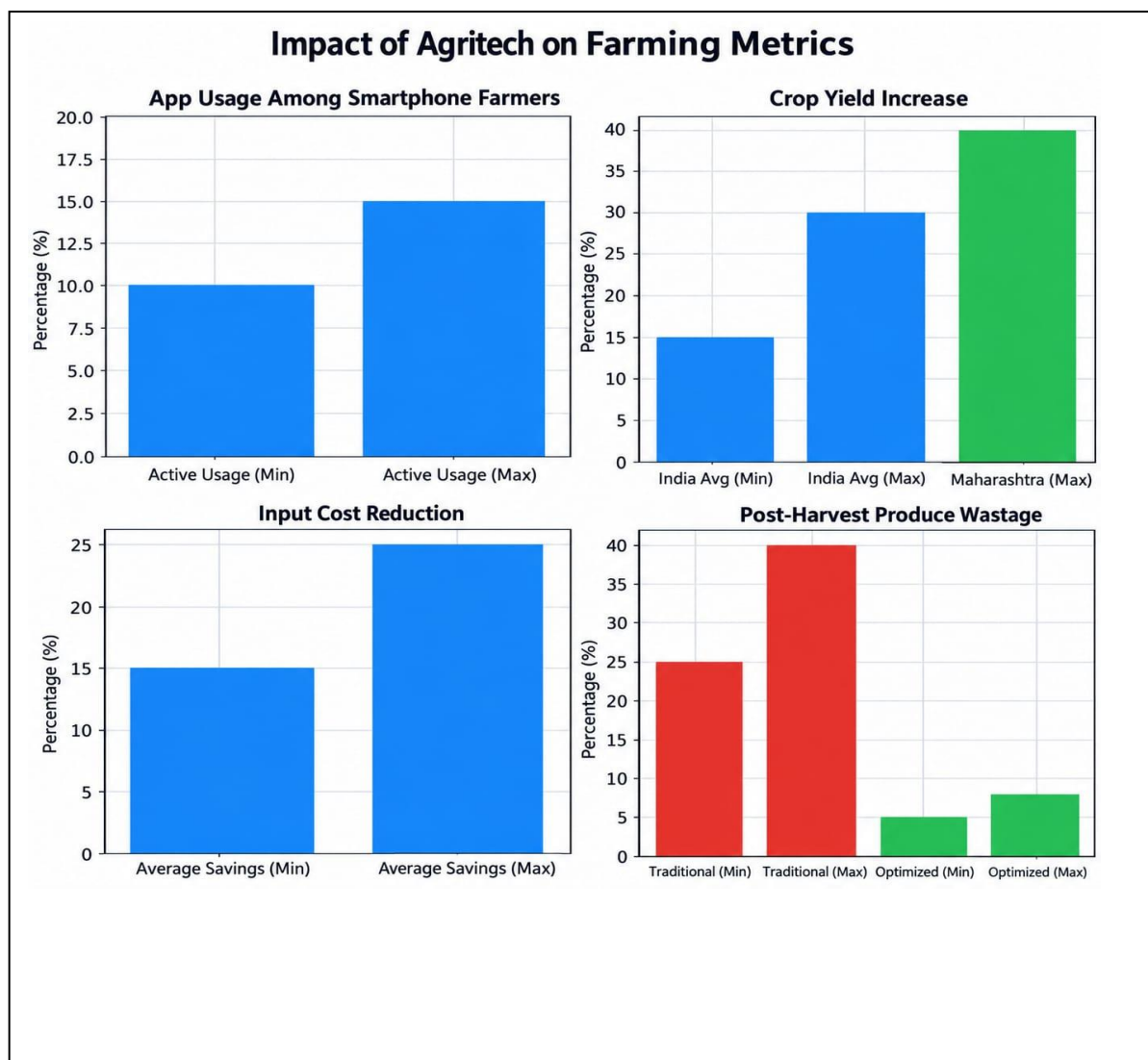
- **Yield Increases:** Implementing precision agriculture apps and AI advisory models yields an average 15% to 30% increase in crop production. In localized Maharashtra pilots, particularly for cash crops like sugarcane and horticulture, yield gains have reached up to 40% per acre.
- **Input Cost Reduction:** Precision farming apps that optimize fertilizer, pesticide, and water application lower overall farm input costs by 15% to 25%.
- **Income Uplift:** Industry research from bodies like EY India and Bain & Company indicates that farm incomes can improve by 20% to 35% when combining precision advisory with direct market linkage platforms.

Land, Crop & Supply Chain Wastage Reduction

- **Soil Degradation & Overuse:** Precision management tools cut excess chemical runoff and over-irrigation by 20% to 30%, which directly mitigates farm soil fatigue and preserves arable land quality.
- **Post-Harvest Wastage:** Traditional post-harvest supply chain losses for perishables (fruits and vegetables) in India range from 25% to 40%. Output-linkage apps (such as DeHaat and Ninjacart) shorten the distribution chain from farm to retailer, reducing produce wastage down to between 5% and 8%.
- **Harvest-Loss Prevention:** Weather alerts and pest early-warning systems prevent catastrophic crop damage, saving an estimated 10% to 15% of cultivable output during erratic climate events.

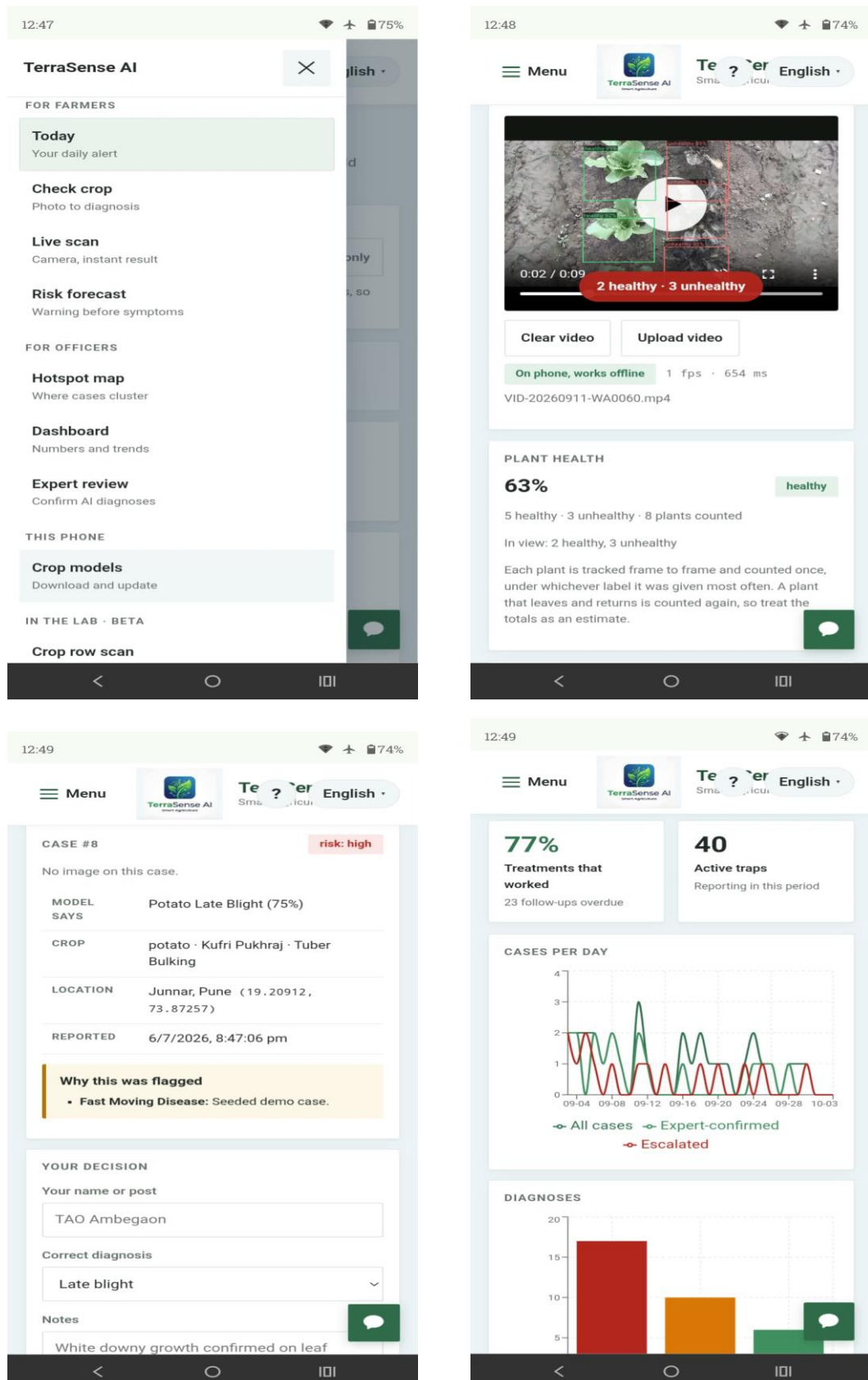
Summary Metrics (Industry Benchmark vs. Maharashtra Context)

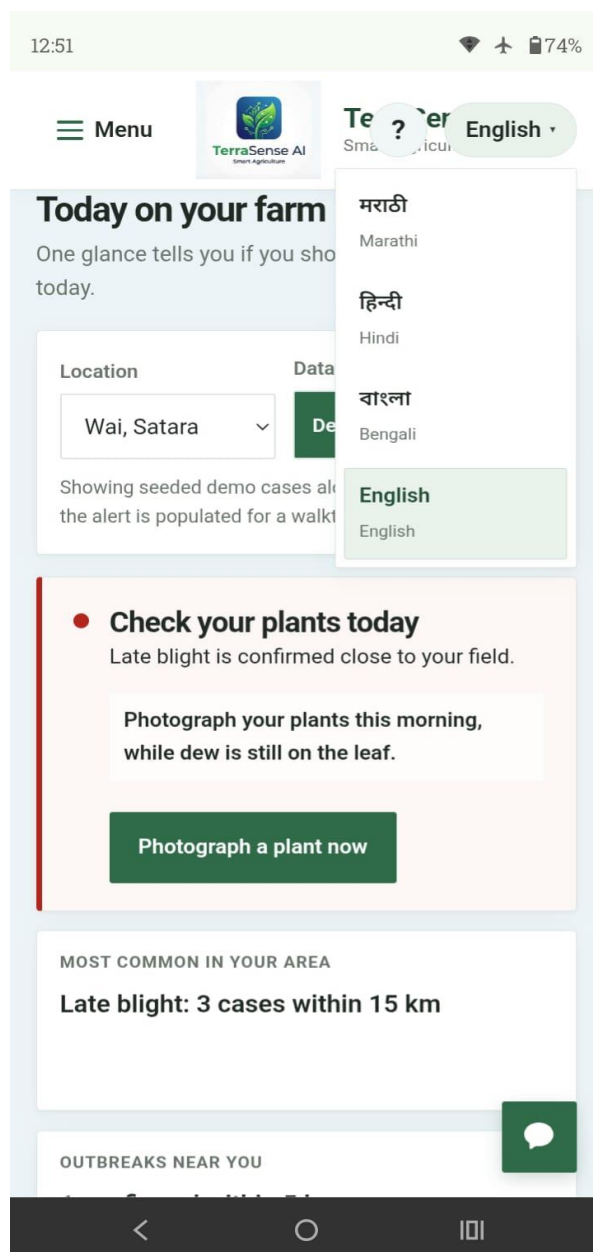
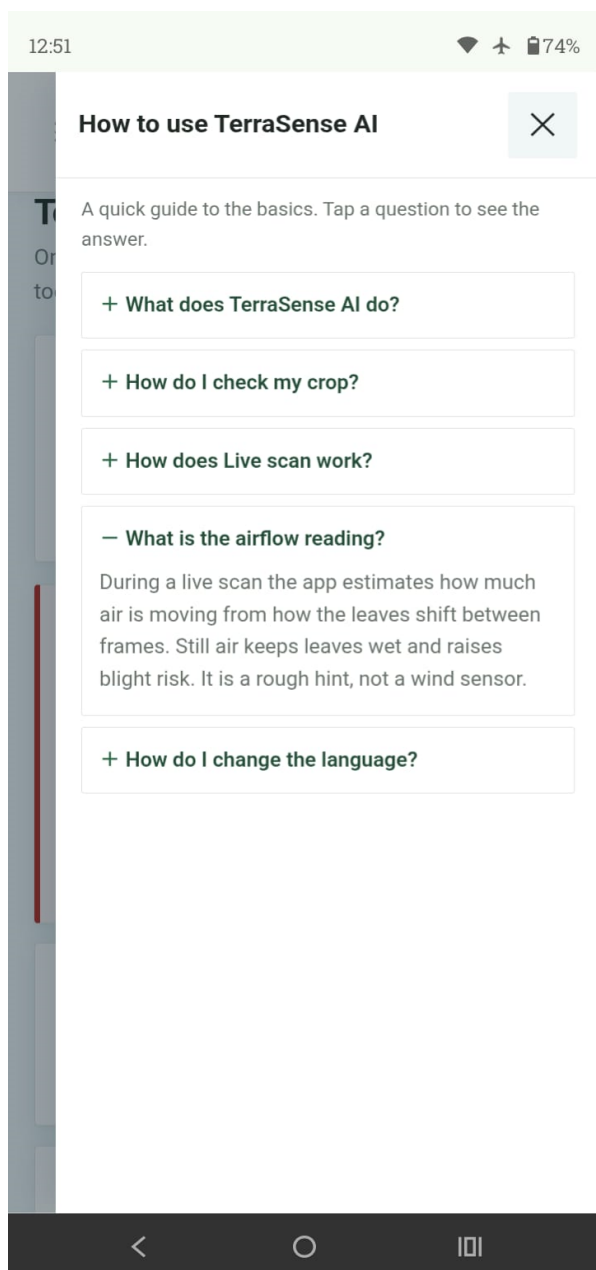
- **Active App Usage:** 10%–15% of smartphone farmers generally, compared to 3 million+ active on MahaVISTAAR / AgriStack in Maharashtra.
- **Crop Yield Boost:** +15% to +30% on average, with up to +40% localized gains in Maharashtra's sugarcane and horticulture.
- **Input Cost Savings:** 15%–25% reduction on average, with substantial savings driven by drone and IoT advisories in Maharashtra.
- **Supply Chain Produce Wastage:** Reduced from ~30% down to 5%–8%, with high impact noted across Western Maharashtra corridors.



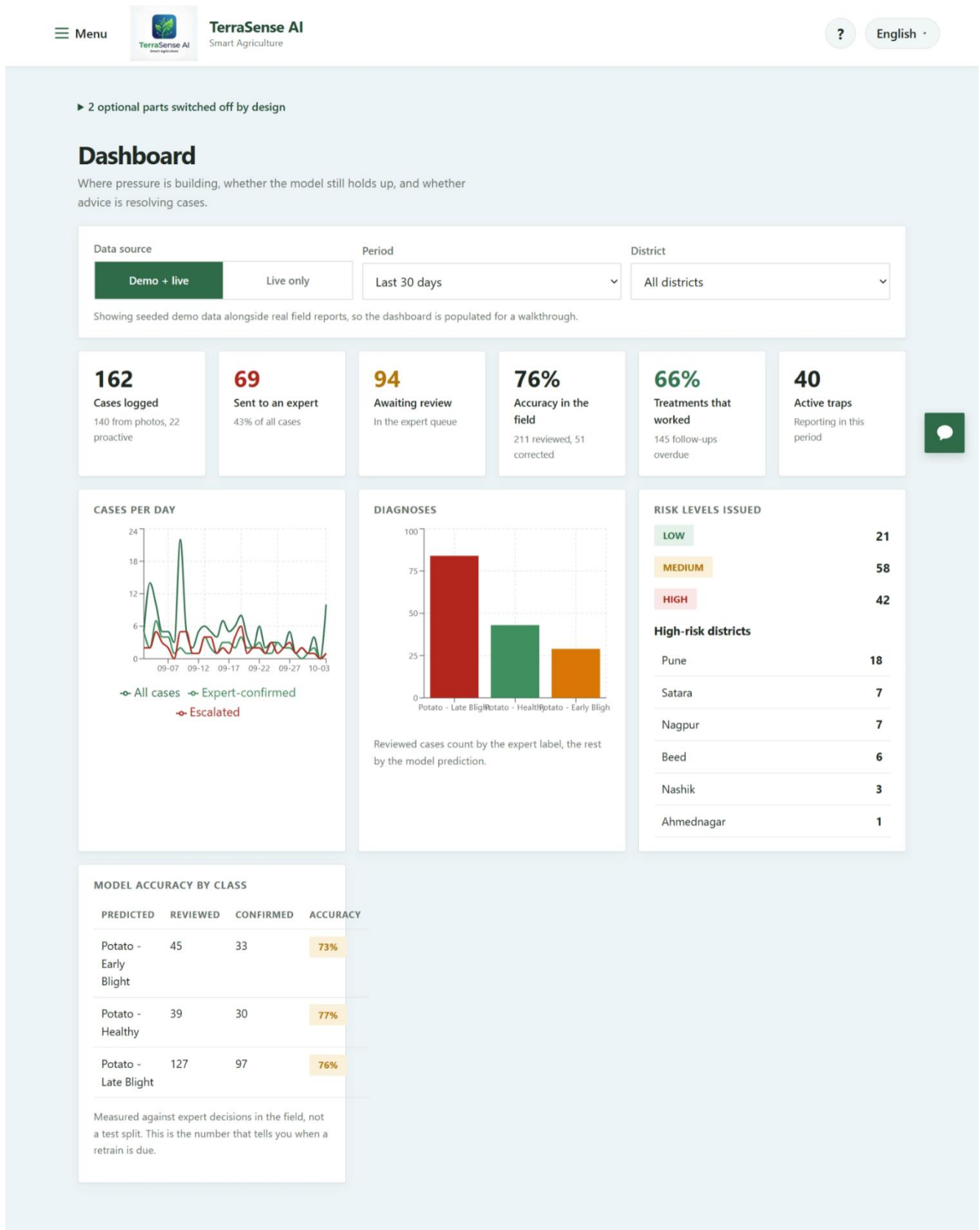
12. Screenshots of our Prototype

12.1 Some screenshots of our mobile app:





12.2 Screenshots of our web app:



Today on your farm

One glance tells you if you should walk your field today.

Location

Manchar, Pune

Data

Demo + live

Live only

Showing seeded demo cases alongside real reports, so the alert is populated for a walkthrough.

• Check your plants today

The weather is right for late blight to start.

Photograph your plants this morning, while dew is still on the leaf.

[Photograph a plant now](#)

MOST COMMON IN YOUR AREA

Late blight: 5 cases within 15 km

OUTBREAKS NEAR YOU

1 confirmed within 5 km

nearest 2.5 km away

WEATHER NOW

24°C, 61% humidity

Early data. This alert gets sharper as more weather and local reports build up.

Field tools

Go straight to a task.

[Scan a crop](#)[View risk forecast](#)[Open field map](#)[Open dashboard](#)

This is a warning to look, not a diagnosis. Photograph a plant to be sure.

[Show alerts for my field](#)

► 2 optional parts switched off by design

Hotspot map

Cases as a live density surface. A confirmed case counts fully, an unreviewed one counts at 0.4, so a spike still shows without sending staff out on unchecked AI output.

View

[Heatmap](#)[Grid](#)

Data source

[Demo + live](#)[Live only](#)

Period

[Last 30 days](#)

Disease

[All](#)☒ Show pest traps

● Low ● Moderate ● High ● Severe 50 confirmed · 39 pending

